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About the UltraWave 12G MWSource : Features

Features

The MWSource generates high-accuracy, low-noise, and low-distortion sine waves. It performs automatic leveling (amplitude level calibration) at each programmed frequency. In addition, the

MWSource controls the Reference DUT LO and DUT Clock. The Reference Synthesizer can drive either the DUT LO or DUT Clock, but not both at the same time.

CW Source

Basic functional specifications are:

•	Frequency range: 50MHz to 12GHz
•	Amplitude range:
–	50MHz to 2.5GHz: -130dBm to +13dBm
–	2.5GHz to 6GHz: -130dBm to +10dBm
–	Above 6GHz: -110dBm to 0dBm
•	Amplitude resolution: 0.1dB

DUT LO

Basic functional specifications are:

•	4:1 splitter with independent output relays
•	Frequency range: 50MHz to 6GHz
•	Amplitude range: -30dBm to +10dBm

DUT Clock

Basic functional specifications are:

•	4:1 splitter with independent output relays
•	Frequency range: 8MHz to 100MHz
•	Frequency resolution: <1Hz

- [Amplitude range: -20 dBm to +13 dBm](#)
-
- [Amplitude resolution: <0.25dB](#)

For UltraWave 12G hardware specifications, see [UltraWave 12G Specifications](#).

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About the UltraWave 12G MWSource : Safety Information

Safety Information

When operating the Microwave Source instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

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About the UltraWave 12G MWSource : Key Terms

Key Terms
See [Key Terms](#) in About the UltraWave 12G Subsystem.

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About the UltraWave 12G MWSource

About the UltraWave 12G MWSource

The MWSource is a programmable CW source that generates signals in the frequency range from 50MHz to 12GHz.

The MWSource logical instrument is also the programming interface for the DUT LO and DUT Clock on a Synthesizer Reference board.

Applications for the MWSource include prescalers, RF amplifiers, TV tuners, cellular and connectivity radios, IF devices, and operational amplifiers.

In addition to the MWSource, the UltraWave 12G subsystem supports several other types of sources:

MWModulatedSource	Source modulated by an arbitrary waveform. See About the UltraWave 12G MWModulatedSource.
MWMultitoneSource	Multitone source for two or more tones. See About the UltraWave 12G MWMultiToneSource.
MWNoiseSource	Noise source with calibrated ENR. See About the UltraWave 12G MWNoiseSource.

This section includes the following topics:

- [Features](#)
- [Safety Information](#)

- [Key Terms](#)

MW Source information includes the following sections:

- [UltraWave 12G MWSource Theory of Operation](#)
- [UltraWave 12G MWSource Programming](#)
- [UltraWave 12G MWSource Debug Display](#)

Related Topics

- [UltraWave 12G Specifications](#)
- [MWSource](#) (VBT syntax)
- [UltraWave 12G DIB Design Guidelines](#)
- [About the UltraWave 12G Subsystem](#)
- [UltraWave 12G Examples](#)

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UltraWave 12G MWSource Debug Display

UltraWave 12G MWSource Debug Display

The UltraWave 12G Microwave Debug Display can display one or more instrument blocks for the MWSource logical instrument, each showing the state of a MWSource in the context of a site. This section includes the following topic:

- [UltraWave 12G MWSource Debug Display Controls](#)

For an overview of all displays supporting the UltraWave 12G subsystem, see [UltraWave 12G Displays](#).
For more information on using debug displays, see the following:

- [Using Debug Displays in TDE](#)
- [Producing Code from Debug Displays](#)

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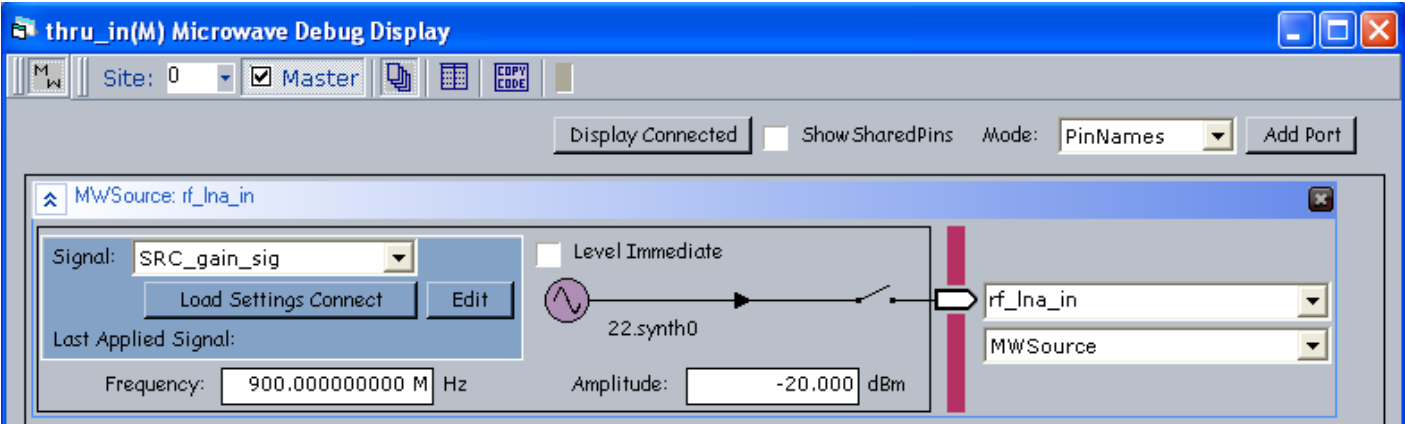


UltraWave 12G MWSrc Debug Display : UltraWave 12G MWSrc Debug Display Controls

UltraWave 12G MWSrc Debug Display Controls

The display controls for the MWSrc assigned to a port allow you to view and set instrument parameters, which are also accessible through the Pins interface of the MWSrc VBT language. The same display is used for DUT LO ports and DUT Clock pins on the MW Reference Synthesizer board.

For information about the common debug display controls, see [UltraWave 12G Displays](#).



MWSrc Logical Instrument Display

The following controls are available on the MWSrc logical instrument block:

Frequency	Displays and sets the source frequency. The tooltip displays the allowable frequency range.
Amplitude	Displays and sets the source amplitude. The tooltip displays the allowable amplitude range.
Level Immediate	Sets the MWSrc leveling mode to Level Immediate. See MWSrc.Pins.LevelImmediate in the VBT Syntax Reference.

synth label	Identifies the synthesizer (slot and number) associated with this Microwave Source logical instrument.
Signal	Name of the Signal used by the source. The available Signals appear on the drop-down list. The MWSource can operate without a Signal.
Load Settings Connect	Loads the selected signal and connects the MWSource.
Info	Displays information for the selected Signal in a pop-up balloon window.
Last Applied Signal	Name of the last Signal applied to the source.
Connection relay	The relay control connects or disconnects the MWSource logical instrument to the specified pin or channel. Click the relay to open or close the connection.

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UltraWave 12G MWSrc Programming

UltraWave 12G MWSrc Programming

The MWSrc is a MW logical instrument in the IG-XL environment. In general, you program the MWSrc like the other MW instruments.

The following topics are available:

- | | |
|---|---|
| • | Sourcing Sine Waves Using the MWSrc |
| • | MWSrc Pattern Control |

Before writing a program using the MWSrc, see [UltraWave 12G Instrument Programming](#). For concepts, a procedure, and an example, see [Sourcing Sine Waves Using the MWSrc](#). For more information, see:

- | | |
|---|--|
| • | MWSrc (VBT Syntax Reference) |
| • | UltraWave 12G Examples |

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UltraWave 12G MWSrc Programming : Sourcing Sine Waves Using the MWSrc

Sourcing Sine Waves Using the MWSrc
This section includes the following topics:

- | | |
|---|--------------------------------------|
| • | Programming Concepts |
| • | Programming Steps |
| • | Example |

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UltraWave 12G MWSrc Programming : Sourcing Sine Waves Using the MWSrc : Programming Concepts

Programming Concepts

Source Leveling

Source level calibration, also called leveling, finely calibrates the MWSrc at each different frequency set in a program. Leveling takes place at run time.

Leveling occurs the first time a specific frequency is set for the MW Source. Later VBT statements setting the same frequency use cached data and do not level, unless leveling is invoked using the Pins.LevelImmediate VBT property for the source type. For example, see the [LevelImmediate](#) property for the Pins interface of TheHdw.MWSrc.

When you call a MWSrc from a pattern, level calibration occurs for all parameter value set combinations before the pattern starts.

MWSrc Reset Values

The table lists the local reset values for the MWSrc.

MWSrc Reset Values	
Setting	Reset Value
Amplitude	-30dBm
Frequency	1.0GHz
Connections	Disconnected

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UltraWave 12G MWSrc Programming : Sourcing Sine Waves Using the MWSrc : Programming Steps

Programming Steps

The MWSrc is a simple CW source instrument producing a sine wave at a programmed frequency and amplitude.

To source a sine wave to an assigned RF port:

- | | |
|----|------------------------------------|
| 1. | Set source signal amplitude. |
| 2. | Set source signal frequency. |
| 3. | Connect to the DUT pin or channel. |

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UltraWave 12G MWSrc Programming : Sourcing Sine Waves Using the MWSrc : Example

Example

```
Function MWSrcSignal(ByVal RF_Frequency As Double, _
                    ByVal input_power As Double, _
                    ByVal Source_Pin As String)

' Set up Source
TheHdw.MWSrc.Pins(Source_Pin).Frequency = RF_Frequency
TheHdw.MWSrc.Pins(Source_Pin).Amplitude = input_power

' Connect MWSrc
TheHdw.MWSrc.Pins(Source_Pin).Connect

' Disconnect MWSrc
TheHdw.MWSrc.Pins(Source_Pin).Disconnect

End Function
```

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UltraWave 12G MWSrc Programming : MWSrc Pattern Control

MWSrc Pattern Control

You can control an MWSrc instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming. You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution. The following topics are available:

- MWSrc Pattern Microcodes
- MWSrc ASCII Pattern Syntax

Note:

To use pattern microcodes to control an MWSrc on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraWave 12G MWSrc Programming : MWSrc Pattern Control : MWSrc Pattern Microcodes

MWSrc Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an MWSrc instrument from a pattern.

MWSrc Pattern Microcodes	
Microcode	Functional Description
LoadSettingsConnect <SignalName>	Instructs an MWSrc to load the settings for the named capture signal and connects the hardware to the DUT. SignalName is required.
Disconnect	Instructs an MWSrc to reset all connections to their disconnected state.

Note the following when using microcodes with an MWSrc:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:

– LoadSettingsConnect: 200 - 400s (typical)

– Disconnect: <200s (typical)

- An MWSource does not use PSets. It provides the equivalent capability using Signals programming in VBT.

- When using an UltraWave logical instrument (such as an MWSource), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.

– IG-XL returns an error if you do not follow this guideline.

– Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see [Using UltraWave 12G Instruments with POP](#).

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[UltraWave 12G MWSrc Programming : MWSrc Pattern Control](#) : MWSrc ASCII Pattern Syntax

MWSrc ASCII Pattern Syntax

MWSrc pattern microcodes are instructions you can place on an individual vector of a pattern. For that vector, a microcode instructs the MWSrc to perform a particular action, such as starting a new signal.

You can only use MWSrc microcodes with pins that the instruments statement has specified as MWSrc. For example:

```
instruments = {
    DUT_AIN:MWSrc;
}
```

For more information, see [Instruments Statement](#) in the [Pattern Language](#) section.

If a vector contains any MWSrc related microcode other than nop, that vector is placed in SRM. The format for specifying an MWSrc microcode on a particular vector is:

(pin-list : MWSrc = microcode)

The following code shows an example of ASCII pattern microcode syntax for an MWSrc. The figure shows this pattern in PatternTool.

```
instruments = {
    Source:MWSrc 1;
}
```

vm_vector

vm_MWSrcOnly_A (\$tset , Dig1)

{

start_label MWSrcOnly:

> BasicSource 0 ;

[illegible]

[illegible]

}

MWSourceOnly.PAT (vm_MWSourceOnly_A - VM)

TSet	Command	Label	Vector	Source (MWSource)	Signal	Comment
BasicSource		start_label MWSourceOnly:	0		0	
BasicSource			1	LoadSettingsConnect GSM	0	
BasicSource	repeat 10000		2		0	1mS wait time
BasicSource			3	LoadSettingsConnect WCDMA	0	
BasicSource	repeat 10000		4		0	1mS wait time
BasicSource			5	Disconnect	0	
BasicSource	repeat 2000		6		0	200uS wait time
BasicSource			7		0	
BasicSource			8		0	
BasicSource			9		0	
BasicSource			10		0	
BasicSource			11		0	
BasicSource			12		0	
BasicSource			13		0	
BasicSource			14		0	
BasicSource			15		0	

Vertical: 65 Vectors | Vectors 0 to 15

MWSource Pattern Microcodes Example

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UltraWave 12G MWSource Theory of Operation

UltraWave 12G MWSource Theory of Operation

The MWSource instrument produces a programmable CW signal using the sine wave output of an RF frequency synthesizer.

In addition, the MWSource logical instrument controls the DUT LO and DUT Clock outputs on a Synthesizer Reference board.

The following topics are available:

- [CW Source Basic Operation](#)
- [DUT LO and DUT Clock Operation](#)
- [MWSource Hardware Functions and Signal Flows](#)

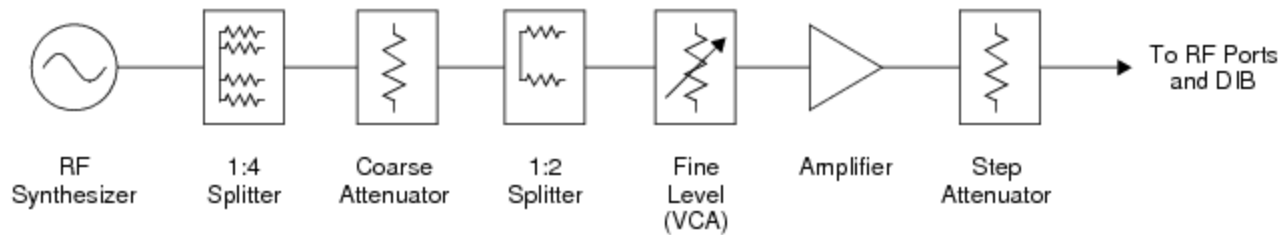
For information on MWSource instrument performance specifications and standards, see [UltraWave 12G Specifications](#).

CW Source Basic Operation

The RF synthesizer used by a MWSource generates a sine wave. The sine wave is routed through a 1:4 splitter to the coarse attenuator and 1:2 splitter feeding the source paths on two front end modules on a Measure board.

The frequency synthesizer is locked to a master clock reference to maintain a coherent timing relationship with the test system.

A coarse attenuator and a front end module attenuate and amplify the signal as needed to set the programmed power level at the DIB. See the figure.



UltraWave 12G MWSource Block Diagram

The front end modules use attenuators for coarse level adjustment, and VCAs to provide fine level adjustment. The signal path depends on the programmed amplitude. The noise level changes when a coarse or step attenuator adjustment occurs due to an amplitude change.

✓ **Note:**

The signal path of a MWSource has more than one separate attenuator and amplifier. All level control elements are combined in the blocks in the diagram.

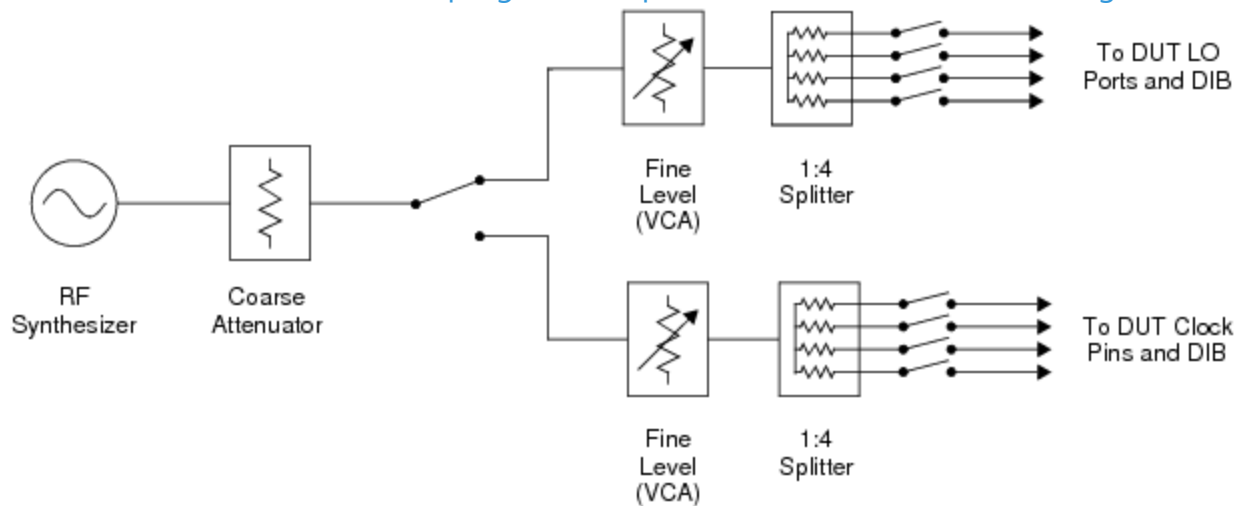
The splitters in the signal path are part of the fanout architecture of the UltraWave 12G. They allow multiple MWSources to connect to multiple RF ports and device pins, without splitters on the DIB. However, a MWSource shares a synthesizer with at least one other MWSource (through a 1:2 splitter) or more than one other MWSource on any other Measure boards (through a 1:4 splitter).

DUT LO and DUT Clock Operation

The RF synthesizer used by a MWSource controlling a DUT LO or DUT Clock output generates a sine wave. The sine wave is routed to either the DUT LO or DUT Clock section on the Reference Synthesizer board. Each section has level control and four independent outputs with disconnect relays.

The frequency synthesizer is locked to a master clock reference to maintain a coherent timing relationship with the test system.

An attenuator and VCA set the programmed power level at the DIB. See the figure.



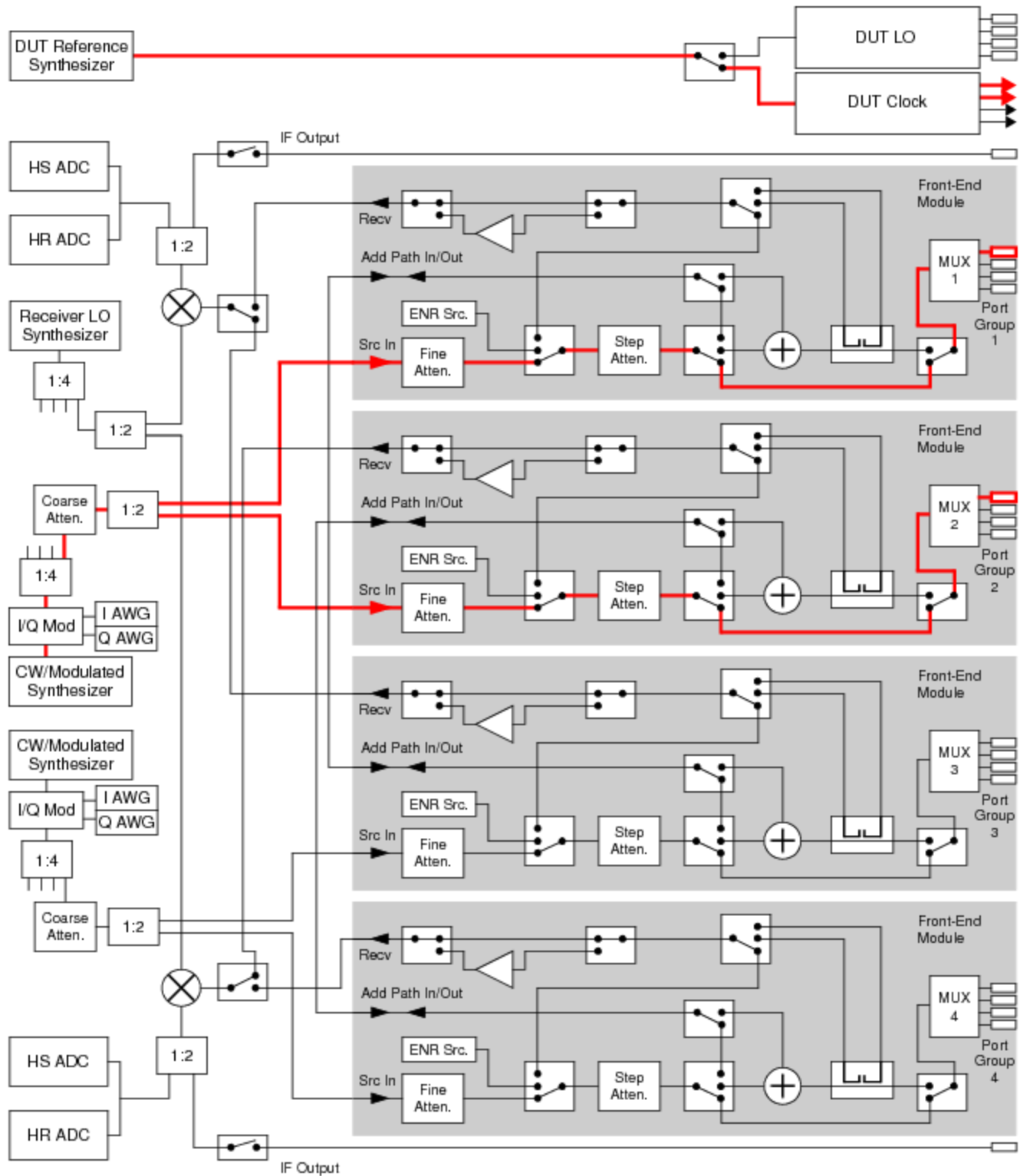
UltraWave 12G DUT LO and DUT Clock Block Diagram

MWSource Hardware Functions and Signal Flows

The figure shows the source signal flows for the connection and operation of two MWSources providing CW signals, and one MWSource providing two DUT Clock signals. Heavy lines represent source signal flows.

The two MWSources share the same RF synthesizer. They must be set to have the same amplitude and frequency settings.

In the figure, only two of the four DUT Clock outputs are connected. Because they share the same RF synthesizer, they must have the same amplitude and frequency settings.



MWSource Hardware Functions and Signal Paths

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